R Notebook

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   4.0.0     v stringr   1.5.1
## v lubridate 1.9.4     v tibble    3.3.0
## v purrr     1.0.4     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(purrr)
library(ggplot2)
library(vegan)
```

```
## Loading required package: permute
```

```r
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(stringr)
library(patchwork)
library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'
##
## The following object is masked from 'package:dplyr':
##
##     group_rows
```

```r
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```r
knitr::opts_chunk$set(dev = "cairo_pdf")
source("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/helper_utils.r")
```

Sample Taxon Reports

```r
# List all CSV files in the directory
csv_files <- list.files(path = "/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/experiment_al
seq_metrics = read.csv("/Users/mattparker/Dropbox/Workspace/moops/manuscript1/data/sequencing_summary_me

# Process all CSV files and combine them
all_data <- bind_rows(lapply(csv_files, process_csv_file))

# separate slurry from pond samples
pond_samples = c("s9","s10","s19","s20")

# Create a metadata dataframe to merge with taxon data
sample_metadata = seq_metrics %>%
  select(sample_name, water_control, infection_class) %>%
  mutate(
  # Library prep: MSSPE if ends with "_M"
  library_prep = if_else(str_ends(sample_name, "_M"), "MSSPE", "NGS"),

  # Remove the _M suffix to get the core sample identifier
  core = str_remove(sample_name, "_M$"),
  # Detect experiment 3 first (Cxxx_LPx)
  is_exp3 = str_detect(core, "C\\d+.*LP\\d+"),

  # Base: first sample ID (e.g. S10, PC1)
  base = str_extract(core, "^[A-Za-z]+\\d+"),

  # Append -2 or -3 only if not exp3
  suffix = case_when(
    !is_exp3 & str_detect(core, "_2$") ~ "-2",
    !is_exp3 & str_detect(core, "_3$") ~ "-3",
```

```r
    TRUE ~ ""
  ),

  base_sample = tolower(paste0(base, suffix)),

  # group slurry vs environmental samples
  sample_type = if_else(base_sample %in% pond_samples, "environmental", "slurry"),

  experiment = case_when(
    is_exp3 ~ 3,
    str_detect(core, "_3$") ~ 2,
    TRUE ~ 1
  )
) %>%
select(sample_name, base_sample, library_prep, experiment, water_control, infection_class, sample_type)

# Merge the taxonomy data with metadata
merged_data <- all_data %>%
  left_join(sample_metadata, by = "sample_name")

# Filter to include only species level (tax_level == 1) to avoid duplication
all_data_species <- merged_data %>%
  filter(tax_level == 1) %>%
  filter(experiment == 3) %>%  # filter in specific experiments
  filter(water_control == "No") %>%
  filter(infection_class != "Definite") # filter out positive controls
```

```r
species_data_filtered = all_data_species %>%
  filter(nt_rpm >= 1 | nr_rpm >= 1 | nt_count >=3) %>% # positive inclusion criteria
  filter(nt_alignment_length > 50)
```

```r
# specific names that should group together
ent_v = "Enterovirus G"
circo_v = "Circovirus"
sapo_v = paste(c("Sapovirus","Sapporo"), collapse = "|")
tes_v = "Teschovirus"
aich_v = "Aichivirus"
astr_v = paste(c("Porcine astrovirus","Mamastrovirus 3"), collapse = "|")
hpv18 = "Alphapapillomavirus"
sapel_v = "Sapelovirus"
boc_v = paste(c("Porcine bocavirus","Bocaparvovirus"), collapse = "|")

# Virus pattern definitions
patterns <- c(
  EVG      = ent_v,
  PCV       = circo_v,
  PSapoV   = sapo_v,
  PTeV      = tes_v,
  PKV       = aich_v,
  PAstV  = astr_v,
  #HPV18   = hpv18,
  PSapeV      = sapel_v,
  PBoV   = boc_v
```

```
)
```

```
# Extract sample metadata (one row per sample)
sample_meta <- all_data_species %>%
  select(sample_name, library_prep, experiment, sample_type) %>%
  distinct()

# Create wide count matrix: taxa x samples
#count_mat <- species_data_filtered %>%
count_mat = all_data_species %>%
  group_by(sample_name, name) %>%
  summarise(reads = mean(nt_count, na.rm = TRUE), .groups = "drop") %>% # counting reads for each taxa f
  pivot_wider(names_from = sample_name, values_from = reads, values_fill = 0) %>%
  column_to_rownames("name")
```

```
# ---- USER PARAMETERS ----
#depths <- c(5e5, 1e6, 2e6, 3e6, 4e6, 5e6, 7e6, 9e6, 11e6)     # Target depths to simulate
depths = c(1e5, 2e5, 3e5, 4e5, 5e5, 6e5, 7e5, 8e5, 9e5, 1e6, 1.1e6, 1.2e6, 1.3e6, 1.4e6, 1.5e6)
n_reps <- 10                         # Replicates per depth

# Start the simulation!
res <- simulate_downsampling(count_mat, depths, n_reps)
```

```
# ---- SUMMARIZE RESULTS ----
# Diversity summary
div_summary <- res %>%
  group_by(depth) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# By library prep
div_summary_libprep <- res %>%
  group_by(depth, library_prep) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# By sample type
div_summary_sampletype <- res %>%
  group_by(depth, sample_type) %>%
  summarise(
    richness_mean = mean(richness),
    richness_sd = sd(richness),
    shannon_mean = mean(shannon),
```

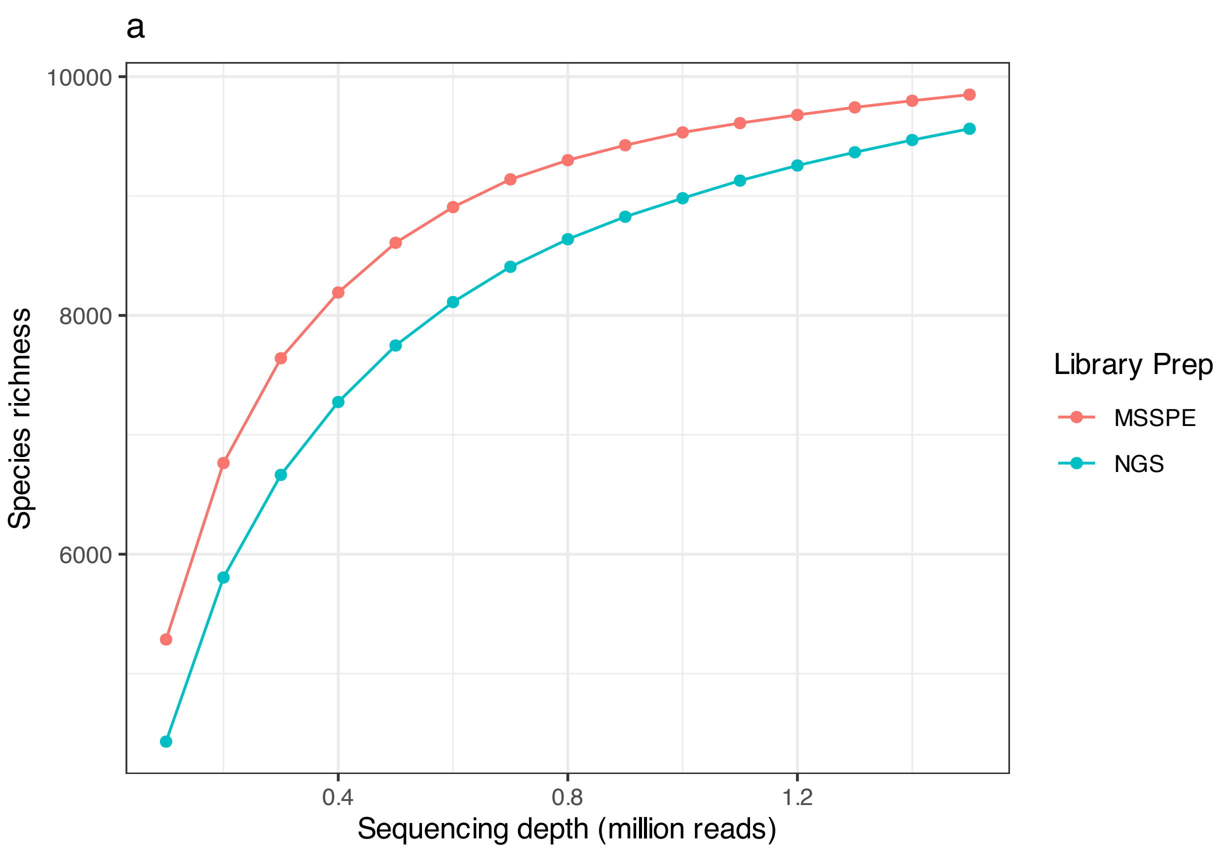
```
    shannon_sd = sd(shannon),
    .groups = "drop"
  )

# remove _M from samples names for comparison
res = res %>%
  mutate(sample = str_remove(sample, "_M.*$"))

res_agg <- res %>%
  rename(richness_val = richness) %>%
  group_by(sample, library_prep, depth) %>%
  summarise(
    richness_mean = mean(richness_val),
    shannon_mean = mean(shannon),
    .groups = "drop"
  )

# do friedman test measuring richness by library prep pairing by sample
friedman_results <- res_agg %>%
  group_by(depth) %>%
  summarise(
    test = list(
      friedman.test(richness_mean ~ library_prep | sample)
    ),
    .groups = "drop"
  ) %>%
  mutate(
    statistic = map_dbl(test, ~ .$statistic),
    p_value   = map_dbl(test, ~ .$p.value)
  )

# Join friedman results back to the overall table
div_summary = div_summary %>%
  left_join(friedman_results, by = "depth")

div_summary = div_summary %>%
  select(-shannon_sd, -richness_sd, -test, -statistic)

div_summary
```

```
## # A tibble: 15 x 4
##      depth richness_mean shannon_mean p_value
##      <dbl>         <dbl>        <dbl>   <dbl>
##  1  100000         4858.         5.87 0.00157
##  2  200000         6284.         5.89 0.00157
##  3  300000         7153.         5.90 0.00157
##  4  400000         7734.         5.90 0.00157
##  5  500000         8178.         5.90 0.0114
##  6  600000         8510.         5.90 0.0114
##  7  700000         8774.         5.91 0.0578
##  8  800000         8969.         5.91 0.0578
##  9  900000         9126.         5.91 0.0578
## 10 1000000         9257.         5.91 0.206
```

```
## 11 1100000         9370.         5.91 0.0578
## 12 1200000         9468.         5.91 0.206
## 13 1300000         9555.         5.91 0.206
## 14 1400000         9634.         5.91 0.206
## 15 1500000         9706.         5.91 0.206
```

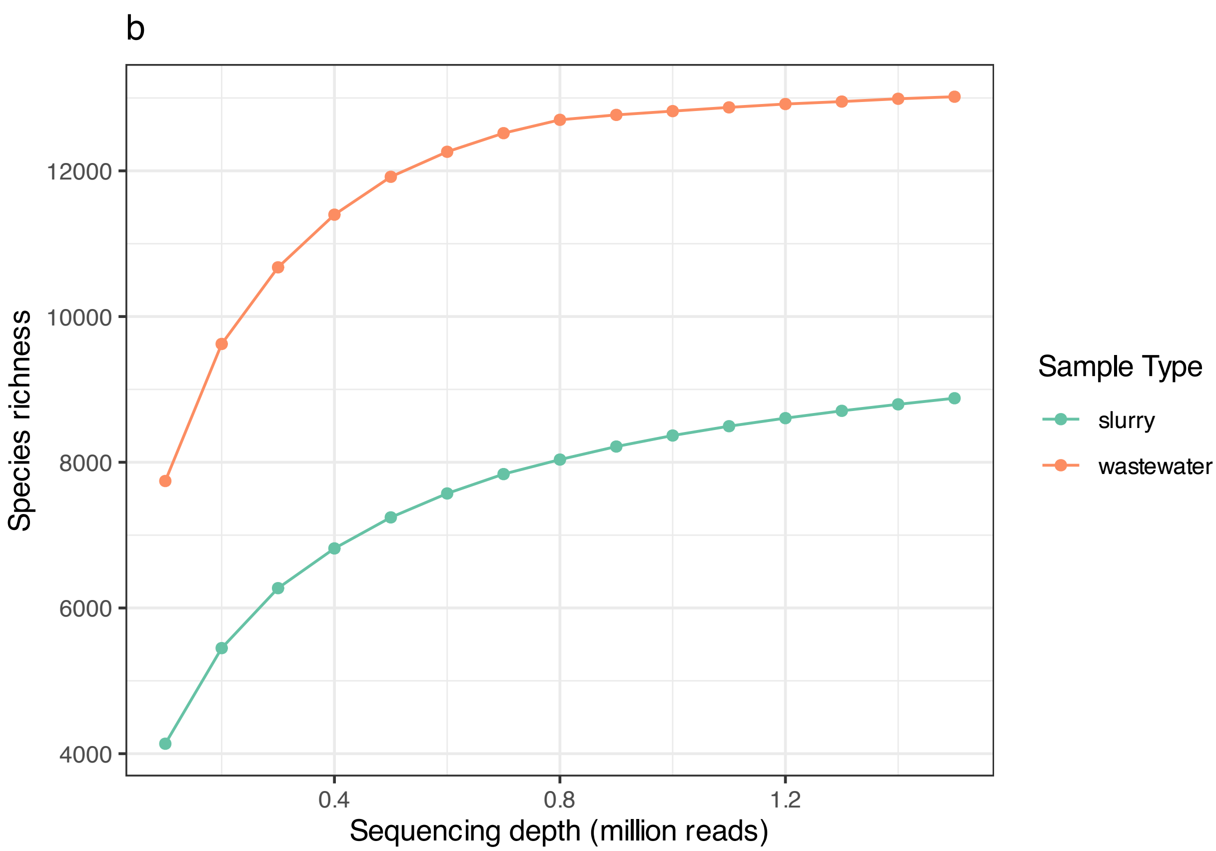
Rarefaction curves

```
div_summary_libprep_filter = div_summary_libprep %>%
  filter(depth <= 7.0e+06) %>%
  rename("Library Prep" = library_prep)

# Plot Rarefaction Curve
p1 <- ggplot(div_summary_libprep_filter, aes(x = depth/1e6, y = richness_mean, fill = `Library Prep`, co
  geom_line(aes(y = richness_mean)) +
  geom_point(aes(y = richness_mean)) +
  #geom_ribbon(aes(ymin = richness_mean - richness_sd, ymax = richness_mean + richness_sd), alpha = 0.2)
  labs(x = "Sequencing depth (million reads)", y = "Species richness",
       title = "a") +
  theme_bw()
p1
```

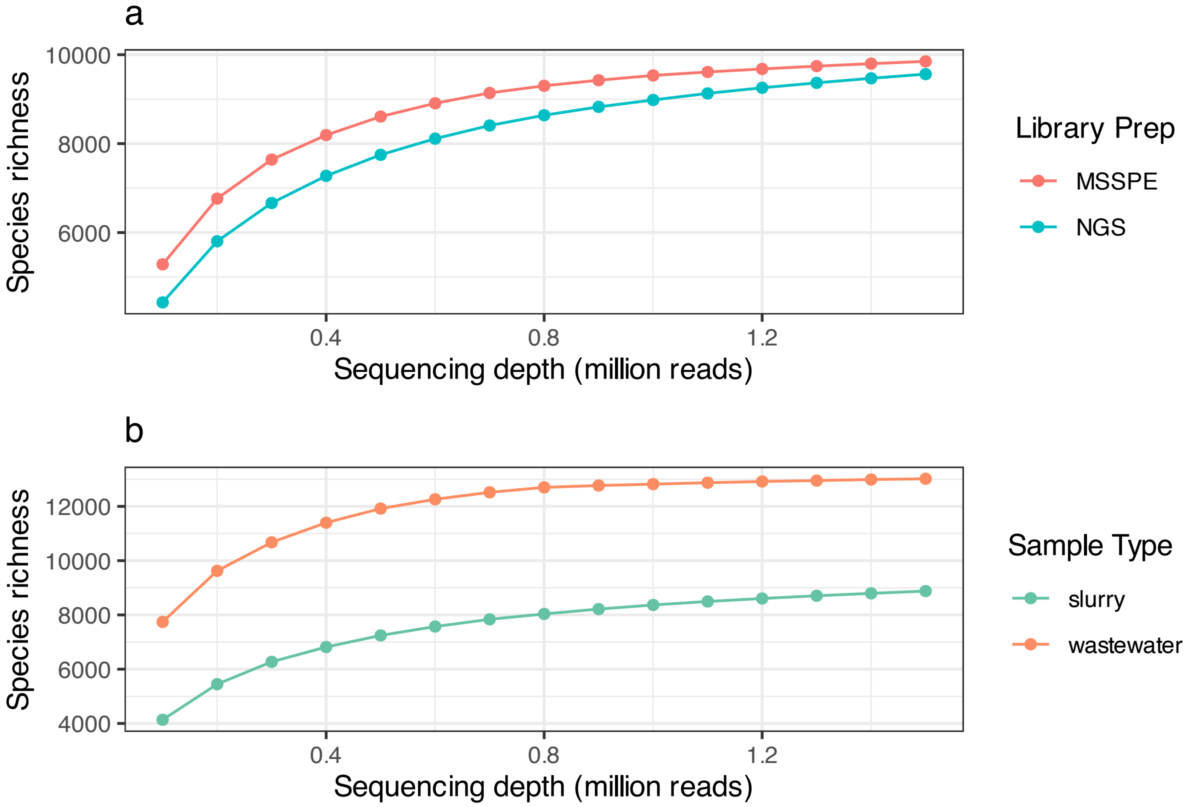



```
div_summary_sampletype_filter = div_summary_sampletype %>%
  filter(depth <= 7.0e+06) %>%
  mutate(
    sample_type = recode(sample_type, `environmental` = "wastewater")
```

```
  ) %>%
  rename("Sample Type" = sample_type)

# Plot Rarefaction Curve
p2 <- ggplot(div_summary_sampletype_filter, aes(x = depth/1e6, y = richness_mean, color = `Sample Type`)
  geom_line(aes(y = richness_mean)) +
  geom_point(aes(y = richness_mean)) +
  scale_color_brewer(palette = "Set2") +
  #geom_ribbon(aes(ymin = richness_mean - richness_sd, ymax = richness_mean + richness_sd), alpha = 0.2
  labs(x = "Sequencing depth (million reads)", y = "Species richness",
       title = "b") +
  theme_bw()
p2
```




```
combined_curves = p1 / p2
combined_curves
```



TARGET-SPECIFIC SEQ DEPTH ANALYSIS

```
# Start the simulation!
res_targets <- simulate_downsampling_targets(
  count_mat,
  depths = c(1e5, 2e5, 3e5, 4e5, 5e5, 6e5, 7e5, 8e5, 9e5, 1e6, 1.1e6, 1.2e6, 1.3e6, 1.4e6, 1.5e6),
  n_reps = 10,
  patterns
)
```

```
# Function to calculate detection probabilities from rarefied reads
summarize_detection <- function(res_targets, detection_threshold) {

  # 1. Pivot longer: each row = sample x replicate x target
  target_cols <- names(res_targets)[
    !(names(res_targets) %in% c("sample", "depth", "rep", "library_prep", "experiment", "sample_type"))
  ]

  det_long <- res_targets %>%
    pivot_longer(
      cols = all_of(target_cols),
      names_to = "target",
      values_to = "total_reads"
    ) %>%
    mutate(
      detected = as.integer(total_reads >= detection_threshold)
    )

  # 2. Detection probability per depth and target
```

```
  det_summary <- det_long %>%
    group_by(depth, target) %>%
    summarise(
      mean_total_reads = mean(total_reads, na.rm = TRUE),
      detect_prob = mean(detected, na.rm = TRUE),
      n_reps = n(),
      .groups = "drop"
    )

  # 3. Detection probability by library prep
  det_summary_libprep <- det_long %>%
    group_by(depth, library_prep, target) %>%
    summarise(
      mean_total_reads = mean(total_reads, na.rm = TRUE),
      detect_prob = mean(detected, na.rm = TRUE),
      n_reps = n(),
      .groups = "drop"
    )

  list(
    long = det_long,
    summary = det_summary,
    summary_libprep = det_summary_libprep
  )
}

# Summarized detection probabilities from simulation
res_summary <- summarize_detection(res_targets, detection_threshold = 5)

# Access outputs
det_long <- res_summary$long                # replicate-level total reads + detected 0/1
det_summary <- res_summary$summary          # mean reads + detection prob per depth x target
det_summary_libprep <- res_summary$summary_libprep  # mean reads + detection prob per depth x target x
```

Linear Mixed Model for Diversity

```
library(glmmTMB)
```

```
## Warning in check_dep_version(dep_pkg = "TMB"): package version mismatch:
## glmmTMB was built with TMB package version 1.9.17
## Current TMB package version is 1.9.18
## Please re-install glmmTMB from source or restore original 'TMB' package (see '?reinstalling' for more
```

```
library(performance)

# get response data from simulation
model_div_data = res %>%
  mutate(
    depth_scaled = depth / 1e5,          # scale depth
    library_prep = factor(library_prep), # convert to factor
    library_prep = relevel(library_prep, ref = "NGS")  # set NGS as reference
  )
```

```r
# create base model for diversity metrics
model_seq_depth_div <- glmmTMB(
  richness ~ library_prep * depth_scaled * sample_type + (1 | sample),
  data = model_div_data,
  family = nbinom2() # use if Richness

)
# print summary of model performance
summary(model_seq_depth_div)
```

```
##  Family: nbinom2  ( log )
## Formula:
## richness ~ library_prep * depth_scaled * sample_type + (1 | sample)
## Data: model_div_data
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    49564     49624    -24772     49544      2990
##
## Random effects:
##
## Conditional model:
##  Groups Name        Variance Std.Dev.
##  sample (Intercept) 0.04597  0.2144
## Number of obs: 3000, groups:  sample, 10
##
## Dispersion parameter for nbinom2 family (): 75.2
##
## Conditional model:
##                                                 Estimate Std. Error z value
## (Intercept)                                     9.084794   0.152289   59.65
## library_prepMSSPE                               0.181647   0.020396    8.91
## depth_scaled                                    0.031581   0.001601   19.73
## sample_typeslurry                              -0.609015   0.170267   -3.58
## library_prepMSSPE:depth_scaled                 -0.011908   0.002258   -5.27
## library_prepMSSPE:sample_typeslurry            -0.038717   0.022841   -1.70
## depth_scaled:sample_typeslurry                  0.014641   0.001793    8.17
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.004332   0.002529    1.71
##                                                Pr(>|z|)
## (Intercept)                                     < 2e-16 ***
## library_prepMSSPE                               < 2e-16 ***
## depth_scaled                                    < 2e-16 ***
## sample_typeslurry                              0.000348 ***
## library_prepMSSPE:depth_scaled                 1.34e-07 ***
## library_prepMSSPE:sample_typeslurry            0.090070 .
## depth_scaled:sample_typeslurry                 3.18e-16 ***
## library_prepMSSPE:depth_scaled:sample_typeslurry 0.086792 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# or profile CI (more accurate but slower)
exp(confint(model_seq_depth_div, parm="beta_", method="profile"))
```

```
##                                                      2.5 %       97.5 %
```

```
## (Intercept)                                       6346.2145250 1.225822e+04
## library_prepMSSPE                                    1.1521465 1.248117e+00
## depth_scaled                                         1.0288492 1.035330e+00
## sample_typeslurry                                    0.3764153 7.858335e-01
## library_prepMSSPE:depth_scaled                       0.9837943 9.925472e-01
## library_prepMSSPE:sample_typeslurry                  0.9198600 1.006085e+00
## depth_scaled:sample_typeslurry                       1.0111854 1.018322e+00
## library_prepMSSPE:depth_scaled:sample_typeslurry     0.9993691 1.009334e+00
```

```
# overdispersion
overdisp_fun(model_seq_depth_div)
```

```
##        chisq        ratio          rdf            p
## 2736.3164922    0.9151560 2990.0000000    0.9996235
```

Microbial diversity sequencing depth tables

```
cat("\\captionsetup{labelformat=empty}")
```

```
# Fixed effects table
fixed_tbl_div <- summary(model_seq_depth_div)$coefficients$cond %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  rename(Estimate = Estimate, SE = `Std. Error`, z = `z value`, p = `Pr(>|z|)`)

# Add odds ratio and 95% CI
fixed_tbl_div <- fixed_tbl_div %>%
  mutate(
    OR = exp(Estimate),
    CI_lower = exp(Estimate - 1.96*SE),
    CI_upper = exp(Estimate + 1.96*SE),
    Percent_change = (OR - 1) * 100,
    Effect = recode(Effect,
                    `(Intercept)` = "intercept",
                    `library_prepMSSPE` = "Library prep",
                    `depth_scaled` = "Sequencing depth",
                    `sample_typeslurry` = "Sample type",
                    `library_prepMSSPE:depth_scaled` = "Lib prep * Sequencing depth",
                    `library_prepMSSPE:sample_typeslurry` = "Lib prep * Sample type",
                    `depth_scaled:sample_typeslurry` = "Sequencing depth * Sample type",
                    `library_prepMSSPE:depth_scaled:sample_typeslurry` = "Lib Prep * Depth * Sample Typ
                    )
  ) %>%
  rename(
    "Percent change" = Percent_change,
    "CI 25%" = CI_lower,
    "CI 75%" = CI_upper
  )

# Random effects table
# Extract variance components
vc <- VarCorr(model_seq_depth_div)
```

```r
random_tbl_div <- data.frame(
  Random_Effect = c("Virus target", "Sample"),
  Variance = c(vc$cond$target[1,1], vc$cond$sample[1,1]),
  SD = sqrt(c(vc$cond$target[1,1], vc$cond$sample[1,1]))
)

# Combine in a nice list for reporting
list(Fixed_Effects = fixed_tbl_div, Random_Effects = random_tbl_div)
```

$Fixed_Effects Effect Estimate SE z p 1 intercept 9.08479448 0.152289477 59.654775 0.000000e+00 2 Library prep 0.18164683 0.020396369 8.905842 5.298239e-19 3 Sequencing depth 0.03158150 0.001600765 19.729005 1.215379e-86 4 Sample type -0.60901540 0.170267197 -3.576822 3.477972e-04 5 Lib prep * Sequencing depth -0.01190794 0.002258071 -5.273500 1.338461e-07 6 Lib prep * Sample type -0.03871692 0.022841427 -1.695031 9.006961e-02 7 Sequencing depth * Sample type 0.01464118 0.001792878 8.166302 3.179878e-16 8 Lib Prep * Depth * Sample Type 0.00433166 0.002529335 1.712569 8.679190e-02 OR CI 25% CI 75% Percent change 1 8820.1528247 6544.0210857 1.188797e+04 8.819153e+05 2 1.1991906 1.1521963 1.248102e+00 1.991906e+01 3 1.0320855 1.0288524 1.035329e+00 3.208549e+00 4 0.5438861 0.3895593 7.593506e-01 -4.561139e+01 5 0.9881627 0.9837989 9.925458e-01 -1.183732e+00 6 0.9620230 0.9199039 1.006071e+00 -3.797700e+00 7 1.0147489 1.0111893 1.018321e+00 1.474889e+00 8 1.0043411 0.9993744 1.009332e+00 4.341055e-01

$Random_Effects Random_Effect Variance SD 1 Virus target 0.04596635 0.2143976 2 Sample 0.04596635 0.2143976

```r
# Optional: use kable() to print nicely in RMarkdown
kable(fixed_tbl_div, digits = 3, caption = "Fixed effects from GLMM (negative binomial)")
```

Fixed effects from GLMM (negative binomial)

Effect	Estimate	SE	z	p	OR	CI 25%	CI 75%	Percent change
intercept	9.085	0.152	59.655	0.000	8820.153	6544.021	11887.965	881915.282
Library prep	0.182	0.020	8.906	0.000	1.199	1.152	1.248	19.919
Sequencing depth	0.032	0.002	19.729	0.000	1.032	1.029	1.035	3.209
Sample type	-0.609	0.170	-3.577	0.000	0.544	0.390	0.759	-45.611
Lib prep * Sequencing depth	-0.012	0.002	-5.274	0.000	0.988	0.984	0.993	-1.184
Lib prep * Sample type	-0.039	0.023	-1.695	0.090	0.962	0.920	1.006	-3.798
Sequencing depth * Sample type	0.015	0.002	8.166	0.000	1.015	1.011	1.018	1.475
Lib Prep * Depth * Sample Type	0.004	0.003	1.713	0.087	1.004	0.999	1.009	0.434

```r
kable(random_tbl_div, digits = 3, caption = "Random effects (variance components) from GLMM")
```

Random effects (variance components) from GLMM

Random_Effect	Variance	SD
Virus target	0.046	0.214

Random_Effect	Variance	SD
Sample	0.046	0.214

Linear Mixed Model for Targets

```
library(glmmTMB)
library(broom.mixed)
library(performance)

det_long <- det_long %>%
  mutate(
    depth_scaled = depth / 1e5,           # scale depth
    library_prep = factor(library_prep), # convert to factor
    library_prep = relevel(library_prep, ref = "NGS")  # set NGS as reference
  )

# create base model for detection probability
model_seq_depth <- glmmTMB(
  detected ~ library_prep * depth_scaled + (1 | target) + (1 | sample),
  data = det_long,
  family = binomial(link = "logit")
)

# print summary of model performance
summary(model_seq_depth)
```

```
##  Family: binomial  ( logit )
## Formula:          detected ~ library_prep * depth_scaled + (1 | target) + (1 |
##     sample)
## Data: det_long
##
##      AIC       BIC    logLik -2*log(L)  df.resid
##    9755.6    9804.1   -4871.8    9743.6     23994
##
## Random effects:
##
## Conditional model:
##  Groups Name        Variance Std.Dev.
##  target (Intercept) 22.160   4.707
##  sample (Intercept)  8.005   2.829
## Number of obs: 24000, groups:  target, 8; sample, 20
##
## Conditional model:
##                                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)                     1.117194   1.914777   0.583    0.560
## library_prepMSSPE               2.041254   1.270877   1.606    0.108
## depth_scaled                    0.244155   0.009365  26.072  < 2e-16 ***
## library_prepMSSPE:depth_scaled -0.089145   0.012683  -7.028 2.09e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# or profile CI (more accurate but slower)
exp(confint(model_seq_depth, parm="beta_", method="profile"))
```

```
##                                      2.5 %      97.5 %
## (Intercept)                     0.05391763 236.9206758
## library_prepMSSPE               0.56224763 106.1100192
## depth_scaled                    1.25347881   1.3003692
## library_prepMSSPE:depth_scaled  0.89219914   0.9377029
```

```r
# standard error
summary(model_seq_depth)$coefficients$cond
```

```
##                                   Estimate  Std. Error    z value      Pr(>|z|)
## (Intercept)                     1.11719450 1.914777327  0.5834592  5.595842e-01
## library_prepMSSPE               2.04125357 1.270876825  1.6061773  1.082349e-01
## depth_scaled                    0.24415472 0.009364722 26.0717526 7.625602e-150
## library_prepMSSPE:depth_scaled -0.08914475 0.012683476 -7.0284165  2.088905e-12
## attr(,"ddf")
## [1] "asymptotic"
```

```r
# check residuals
r2_model <- r2_nakagawa(model_seq_depth)
r2_model
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.905
##      Marginal R2: 0.035
```

```r
# check overdispersion
overdisp_fun(model_seq_depth)
```

```
##       chisq       ratio         rdf           p
## 50252.78872     2.09439 23994.00000     0.00000
```

```r
# Fit zero-inflated model for comparison
model_zi <- glmmTMB(
  detected ~ library_prep * depth_scaled + (1|target) + (1|sample),
  ziformula = ~1,
  data = det_long,
  family = binomial
)

# Compare AIC
AIC(model_seq_depth, model_zi)
```

```
##                 df      AIC
## model_seq_depth  6 9755.630
## model_zi         7 9679.706
```

```r
# Compare alternative sequencing depth modeling

# try log transformed depth
det_long <- det_long %>%
  mutate(log_depth = log(depth))

model_log <- glmmTMB(
  detected ~ library_prep * log_depth + (1|target) + (1|sample),
  data = det_long,
  family = binomial
)
# print AIC comparison
AIC(model_seq_depth, model_log)
```

```
##                 df     AIC
## model_seq_depth  6 9755.63
## model_log        6 9318.77
```

Viral target sequencing depth tables

```r
cat("\\captionsetup{labelformat=empty}")
```

```r
# Fixed effects table
fixed_tbl <- summary(model_seq_depth)$coefficients$cond %>%
  as.data.frame() %>%
  rownames_to_column("Effect") %>%
  rename(Estimate = Estimate, SE = `Std. Error`, z = `z value`, p = `Pr(>|z|)`)

# Add odds ratio and 95% CI
fixed_tbl <- fixed_tbl %>%
  mutate(
    OR = exp(Estimate),
    CI_lower = exp(Estimate - 1.96*SE),
    CI_upper = exp(Estimate + 1.96*SE),
    Effect = recode(Effect,
                    `(Intercept)` = "intercept",
                    `library_prepMSSPE` = "Library prep",
                    `depth_scaled` = "Sequencing depth",
                    `library_prepMSSPE:depth_scaled` = "Library prep * Sequencing depth"
                    )
  ) %>%
  rename(
    "CI 25%" = CI_lower,
    "CI 75%" = CI_upper
  )

# Random effects table
# Extract variance components
vc <- VarCorr(model_seq_depth)

random_tbl <- data.frame(
  Random_Effect = c("Virus target", "Sample"),
```

```
  Variance = c(vc$cond$target[1,1], vc$cond$sample[1,1]),
  SD = sqrt(c(vc$cond$target[1,1], vc$cond$sample[1,1]))
)

# Combine in a nice list for reporting
list(Fixed_Effects = fixed_tbl, Random_Effects = random_tbl)
```

$Fixed_Effects Effect Estimate SE z 1 intercept 1.11719450 1.914777327 0.5834592 2 Library prep 2.04125357 1.270876825 1.6061773 3 Sequencing depth 0.24415472 0.009364722 26.0717526 4 Library prep * Sequencing depth -0.08914475 0.012683476 -7.0284165 p OR CI 25% CI 75% 1 5.595842e-01 3.0562678 0.07166383 130.3415168 2 1.082349e-01 7.7002559 0.63784179 92.9602649 3 7.625602e-150 1.2765418 1.25332480 1.3001889 4 2.088905e-12 0.9147132 0.89225406 0.9377376

$Random_Effects Random_Effect Variance SD 1 Virus target 22.159628 4.707401 2 Sample 8.004824 2.829280

```
# Optional: use kable() to print nicely in RMarkdown
kable(fixed_tbl, digits = 3, caption = "Fixed effects from GLMM (binomial)")
```

Fixed effects from GLMM (binomial)

Effect	Estimate	SE	z	p	OR	CI 25%	CI 75%
intercept	1.117	1.915	0.583	0.560	3.056	0.072	130.342
Library prep	2.041	1.271	1.606	0.108	7.700	0.638	92.960
Sequencing depth	0.244	0.009	26.072	0.000	1.277	1.253	1.300
Library prep * Sequencing depth	-0.089	0.013	-7.028	0.000	0.915	0.892	0.938

```
kable(random_tbl, digits = 3, caption = "Random effects (variance components) from GLMM")
```

Random effects (variance components) from GLMM

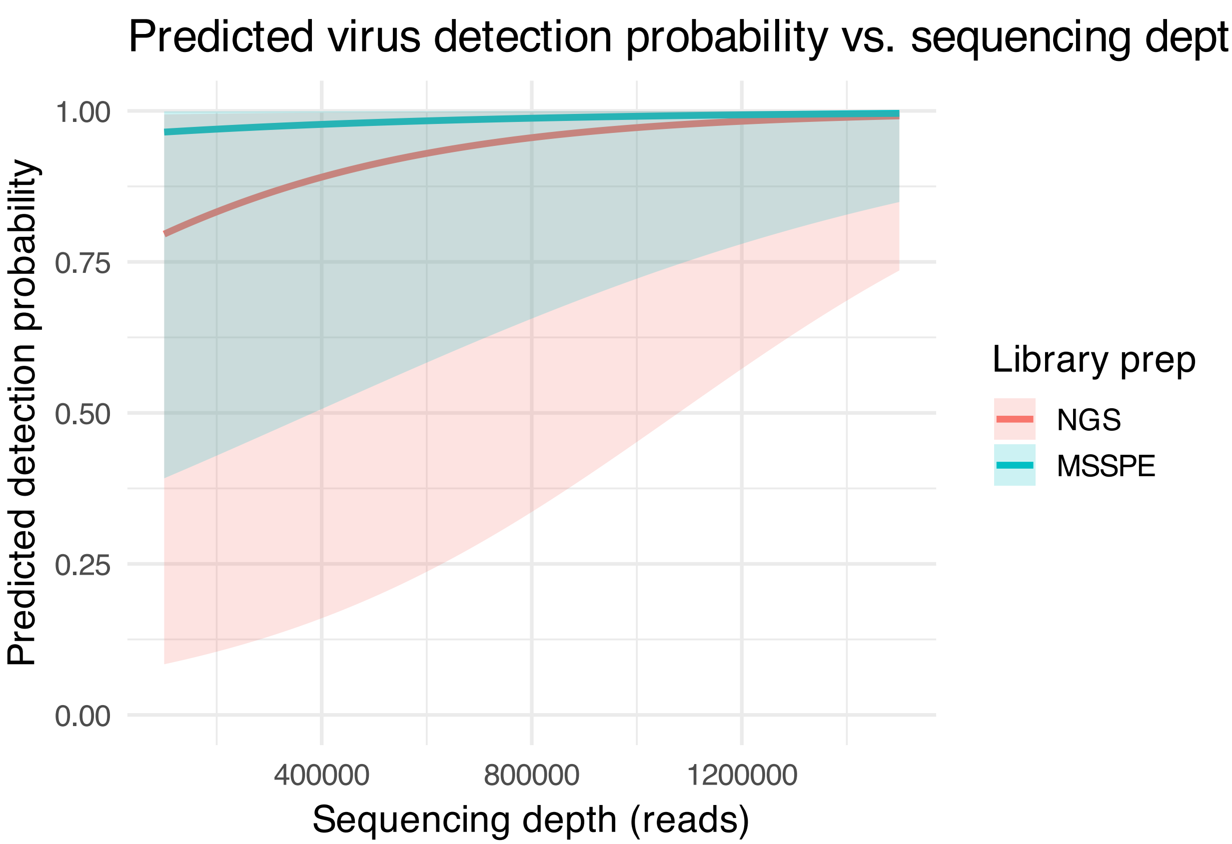
Random_Effect	Variance	SD
Virus target	22.160	4.707
Sample	8.005	2.829

Viral sequencing depth plots

```
# Create a grid of sequencing depths for prediction
depth_grid <- seq(min(det_long$depth_scaled),
                  max(det_long$depth_scaled),
                  length.out = 100)

newdata <- expand.grid(
  library_prep = c("MSSPE", "NGS"),
  depth_scaled = depth_grid
)

# Predicted probabilities and 95% CI using emmeans
emm <- emmeans(model_seq_depth, ~ library_prep | depth_scaled,
```

```r
                at = list(depth_scaled = depth_grid), type="response")

emm_df <- as.data.frame(emm)

# Rename columns for clarity
emm_df <- emm_df %>%
  rename(Detection_Prob = prob,
         CI_Lower = asymp.LCL,
         CI_Upper = asymp.UCL)

# Plot
ggplot(emm_df, aes(x = depth_scaled*1e5, y = Detection_Prob, color = library_prep)) +
  geom_line(size = 1.2) +
  geom_ribbon(aes(ymin = CI_Lower, ymax = CI_Upper, fill = library_prep), alpha = 0.2, color = NA) +
  scale_x_continuous(name = "Sequencing depth (reads)") +
  scale_y_continuous(name = "Predicted detection probability", limits = c(0,1)) +
  labs(color = "Library prep", fill = "Library prep") +
  theme_minimal(base_size = 14) +
  ggtitle("Predicted virus detection probability vs. sequencing depth")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Tables

```r
cat("\\captionsetup{labelformat=empty}")
```

```r
det_summary_libprep_box_format <- det_summary_libprep %>%
  group_by(depth, library_prep, target) %>%
  summarise(mean_prob = mean(detect_prob, na.rm = TRUE),
            .groups = "drop")

det_summary_libprep_box_format
```

A tibble: 240 x 4

```
depth library_prep target mean_prob
<dbl> <chr>        <chr>      <dbl>
```

1 100000 MSSPE EVG 0.4 2 100000 MSSPE PAstV 1
3 100000 MSSPE PBoV 1
4 100000 MSSPE PCV 0.55 5 100000 MSSPE PKV 0.01 6 100000 MSSPE PSapeV 0.55 7 100000 MSSPE PSapoV 0.72 8 100000 MSSPE PTeV 0.5 9 100000 NGS EVG 0.32 10 100000 NGS PAstV 1
i 230 more rows

```r
det_summary_wide <- det_summary_libprep_box_format %>%
  group_by(target, depth, library_prep) %>%               # average across library preps if multiple
  summarise(mean_prob = mean(mean_prob), .groups = "drop") %>%
  mutate(
    depth = paste0(depth / 1e6, "M")
  ) %>%
  pivot_wider(
    names_from = c(depth),
    values_from = mean_prob
  )

det_summary_wide
```

A tibble: 16 x 17

target library_prep `0.1M` `0.2M` `0.3M` `0.4M` `0.5M` `0.6M` `0.7M` `0.8M` 1 EVG MSSPE 0.4 0.67 0.8 0.81 0.87 0.89 0.9 0.9 2 EVG NGS 0.32 0.57 0.62 0.68 0.7 0.74 0.77 0.85 3 PAstV MSSPE 1 1 1 1 1 1 1 1
4 PAstV NGS 1 1 1 1 1 1 1 1
5 PBoV MSSPE 1 1 1 1 1 1 1 1
6 PBoV NGS 0.86 0.93 0.94 1 1 1 1 1
7 PCV MSSPE 0.55 0.68 0.82 0.86 0.9 0.9 0.9 0.9 8 PCV NGS 0.32 0.55 0.63 0.7 0.7 0.74 0.76 0.76 9 PKV MSSPE 0.01 0.03 0.05 0.14 0.19 0.25 0.27 0.28 10 PKV NGS 0 0 0 0.01 0 0.01 0 0
11 PSapeV MSSPE 0.55 0.59 0.6 0.6 0.6 0.6 0.6 0.6 12 PSapeV NGS 0.42 0.6 0.68 0.73 0.77 0.77 0.76 0.8 13 PSapoV MSSPE 0.72 0.85 0.89 0.88 0.9 0.9 0.9 0.9 14 PSapoV NGS 0.48 0.7 0.9 0.94 0.95 1 0.99 1
15 PTeV MSSPE 0.5 0.62 0.68 0.69 0.7 0.7 0.7 0.7 16 PTeV NGS 0.38 0.51 0.54 0.61 0.62 0.67 0.67 0.69 # i 7 more variables: `0.9M` , `1M` , `1.1M` , `1.2M` , # `1.3M` , `1.4M` , `1.5M`

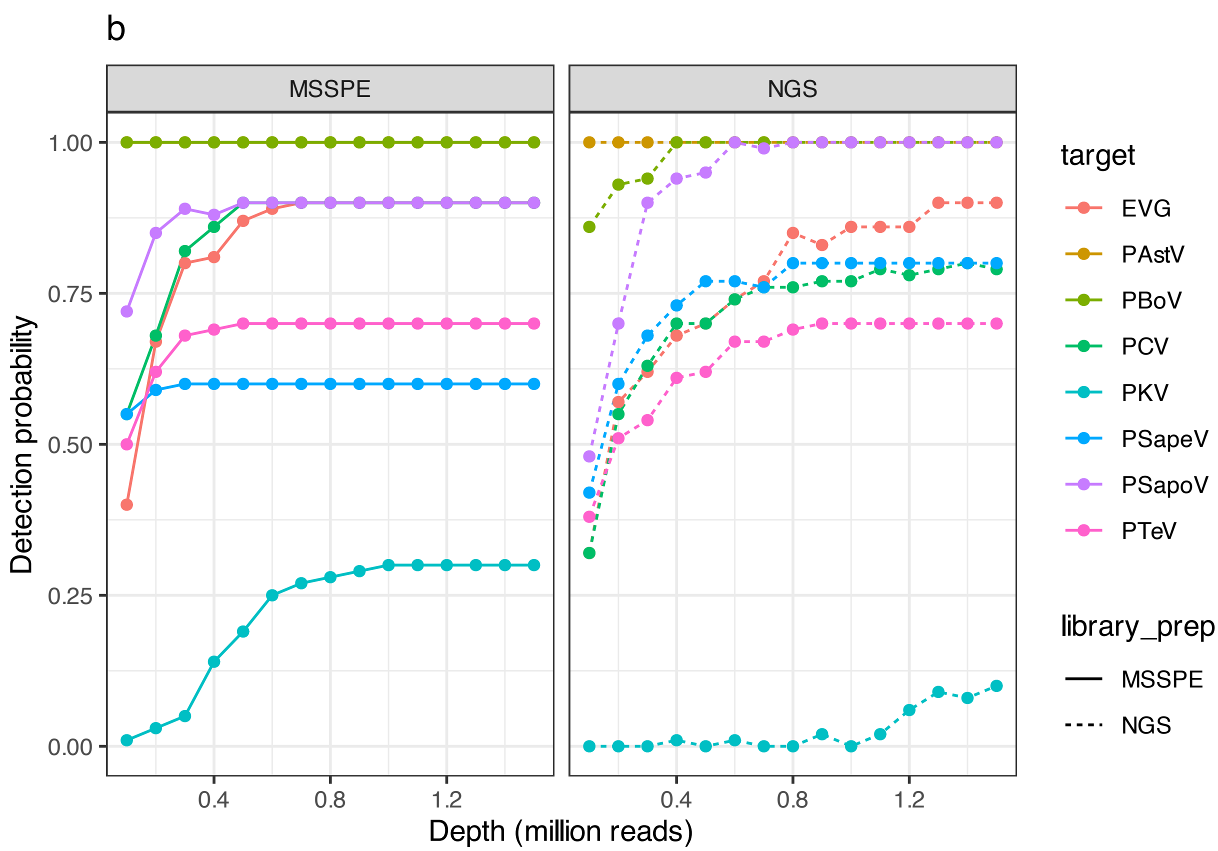
Detection probability per viral target at varying sequencing depths

Virus	Library Prep	0.1M	0.3M	0.5M	0.7M	0.9M	1.1M	1.3M	1.5M
EVG	MSSPE	0.40	0.80	0.87	0.90	0.90	0.90	0.90	0.90
EVG	NGS	0.32	0.62	0.70	0.77	0.83	0.86	0.90	0.90
PAstV	MSSPE	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PAstV	NGS	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PBoV	MSSPE	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
PBoV	NGS	0.86	0.94	1.00	1.00	1.00	1.00	1.00	1.00
PCV	MSSPE	0.55	0.82	0.90	0.90	0.90	0.90	0.90	0.90
PCV	NGS	0.32	0.63	0.70	0.76	0.77	0.79	0.79	0.79
PKV	MSSPE	0.01	0.05	0.19	0.27	0.29	0.30	0.30	0.30
PKV	NGS	0.00	0.00	0.00	0.00	0.02	0.02	0.09	0.10
PSapeV	MSSPE	0.55	0.60	0.60	0.60	0.60	0.60	0.60	0.60
PSapeV	NGS	0.42	0.68	0.77	0.76	0.80	0.80	0.80	0.80
PSapoV	MSSPE	0.72	0.89	0.90	0.90	0.90	0.90	0.90	0.90
PSapoV	NGS	0.48	0.90	0.95	0.99	1.00	1.00	1.00	1.00
PTeV	MSSPE	0.50	0.68	0.70	0.70	0.70	0.70	0.70	0.70
PTeV	NGS	0.38	0.54	0.62	0.67	0.70	0.70	0.70	0.70

```
det_summary_wide %>%
  rename(
    "Virus" = target,
    "Library Prep" = library_prep
  ) %>%
  select(-`0.2M`,-`0.4M`,-`0.6M`,-`0.8M`,-`1M`,-`1.2M`,-`1.4M`) %>%
  kable(
    format = "latex",  # or "latex" if using LaTeX
    caption = "Detection probability per viral target at varying sequencing depths",
    digits = 2,
    format.args = list(big.mark = ",")
  ) %>%
  kable_styling(
    full_width = TRUE,
    position = "center",
  )
```

```
# Plot Target Curve
#rare_targets_curve <- ggplot(det_summary, aes(x = depth/1e6, y = detect_prob, color = target)) +
  #geom_line() + geom_point() +
  #facet_wrap(~threshold, labeller = labeller(threshold = function(x) paste(" ", x, "reads"))) +
  #labs(x = "Depth (million reads)", y = "Detection probability",
       #title = "a") +
  #theme(legend.position = "none")

#rare_targets_curve



# Plot Library Prep Curve
rare_targets_curve_prep <- ggplot(det_summary_libprep_box_format, aes(x = depth/1e6, y = mean_prob, colo
  geom_line() +   # small horizontal dodge to separate lines
```

```
  #geom_point(position = position_jitter(width = 0.3, height = 0.0)) +  # jitter x slightly
  geom_point() +
  facet_wrap(~library_prep) +
  labs(x = "Depth (million reads)", y = "Detection probability",
       title = "b") +
  theme_bw()
rare_targets_curve_prep
```




```
#combined_target_curves = rare_targets_curve / rare_targets_curve_prep
#combined_target_curves
```